```
% ANALISIS CINEMATICO - SISTEMA DE INTERACCIÓN

% Limpieza del entorno de trabajo
clearvars
close all
clc

% Variables simbólicas dependientes del tiempo
syms th1(t) th2(t) th3(t) th4(t) th5(t) t
syms l1 l2 l3 l4 l5

% Tipo de articulaciones (0 = rotacional, 1 = prismática)
RP = [0 0 0 0 0];

% Vector de coordenadas generalizadas
Q = [th1 th2 th3 th4 th5];
disp('Variables articulares del sistema de interacción');
```

```
Variables articulares del sistema de interacción
```

```
pretty(Q)
```

```
(th1(t), th2(t), th3(t), th4(t), th5(t))
```

```

% Vector de velocidades articulares
Qp = diff(Q,t);
disp('Velocidades articulares');
```

```
Velocidades articulares
```

```
pretty(Qp)
```

```
/  d          d          d          d          d         \
| -- th1(t), -- th2(t), -- th3(t), -- th4(t), -- th5(t) |
\ dt         dt         dt         dt         dt        /
```

```

% Grados de libertad
GDL = length(RP);

%% Definición de posiciones y rotaciones por articulación

% Articulación 1
P(:,:,1) = [0; 0; l1];
R(:,:,1) = [ 0,  sin(th1), -cos(th1);
             0,  cos(th1),  sin(th1);
             1,        0,          0];

% Articulación 2
P(:,:,2) = [-l2*sin(th2); l2*cos(th2); 0];
R(:,:,2) = eye(3);

```

```
% Articulación 3
P(:,:,3) = [-l3*sin(th3);  l3*cos(th3); 0];
R(:,:,3) = [ 0, sin(th4),  cos(th4);
             1,        0,          0;
             0, cos(th4), sin(th4)];

% Articulación 4
P(:,:,4) = [0; 0; -l4];
R(:,:,4) = [ cos(th3), 0, -sin(th3);
            -sin(th3), 0, -cos(th3);
                    0, 1,        0];

% Articulación 5
P(:,:,5) = [l5*cos(th5); -l5*sin(th5); 0]; %Si no funciona, cambiar el seno por el
coseno y viceversa
R(:,:,5) = [  cos(th5),  0,  sin(th5);
             -sin(th5),  0,  cos(th5);
                     0, -1,        0];

%% Construcción de matrices homogéneas

Vector_Zeros = zeros(1,3);

A = sym(zeros(4,4,GDL));
T = sym(zeros(4,4,GDL));
PO = sym(zeros(3,1,GDL));
RO = sym(zeros(3,3,GDL));

for i = 1:GDL

    % Matriz homogénea local
    A(:,:,i) = simplify([R(:,:,i) P(:,:,i); Vector_Zeros 1]);
    disp(['Transformación local A',num2str(i)])
    pretty(A(:,:,i))

    % Transformación global acumulada
    if i == 1
        T(:,:,i) = A(:,:,i);
    else
        T(:,:,i) = T(:,:,i-1)*A(:,:,i);
    end

    T(:,:,i) = simplify(T(:,:,i));
    disp(['Transformación global T',num2str(i)])
    pretty(T(:,:,i))

    % Rotación y posición respecto al marco base
    RO(:,:,i) = T(1:3,1:3,i);
    PO(:,:,i) = T(1:3,4,i);
end
```

```
Transformación local A1
/ 0, sin(th1(t)), -cos(th1(t)),  0 \
|                                   |
| 0, cos(th1(t)),  sin(th1(t)),  0 |
|                                   |
| 1,      0,            0,      l1 |
|                                   |
\ 0,      0,            0,       1 /
Transformación global T1
/ 0, sin(th1(t)), -cos(th1(t)),  0 \
|                                   |
| 0, cos(th1(t)),  sin(th1(t)),  0 |
|                                   |
| 1,      0,            0,      l1 |
|                                   |
\ 0,      0,            0,       1 /
Transformación local A2
/ 1, 0, 0, -l2 sin(th2(t)) \
|                           |
| 0, 1, 0,  l2 cos(th2(t)) |
|                           |
| 0, 0, 1,        0        |
|                           |
\ 0, 0, 0,        1        /
Transformación global T2
/ 0, sin(th1(t)), -cos(th1(t)), l2 cos(th2(t)) sin(th1(t)) \
|                                                           |
| 0, cos(th1(t)),  sin(th1(t)), l2 cos(th1(t)) cos(th2(t)) |
|                                                           |
| 1,      0,            0,          l1 - l2 sin(th2(t))    |
|                                                           |
\ 0,      0,            0,                   1             /
Transformación local A3
/ 0, sin(th4(t)), cos(th4(t)), -l3 sin(th3(t)) \
|                                               |
| 1,      0,           0,       l3 cos(th3(t)) |
|                                               |
| 0, cos(th4(t)), sin(th4(t)),        0        |
|                                               |
\ 0,      0,           0,             1        /
Transformación global T3
/ sin(th1(t)), -cos(th1(t)) cos(th4(t)), -cos(th1(t)) sin(th4(t)),           sin(th1(t)) #1           \
|                                                                                                      |
| cos(th1(t)),  cos(th4(t)) sin(th1(t)),  sin(th1(t)) sin(th4(t)),           cos(th1(t)) #1           |
|                                                                                                      |
|      0,             sin(th4(t)),              cos(th4(t)),       l1 - l2 sin(th2(t)) - l3 sin(th3(t)) |
|                                                                                                      |
\      0,                   0,                         0,                              1                 /

where

   #1 == l2 cos(th2(t)) + l3 cos(th3(t))
Transformación local A4
/  cos(th3(t)), 0, -sin(th3(t)),  0  \
|                                     |
| -sin(th3(t)), 0, -cos(th3(t)),  0  |
|                                     |
|       0,      1,       0,      -l4 |
|                                     |
\       0,      0,       0,       1  /
Transformación global T4
/ cos(th3(t)) sin(th1(t)) + cos(th1(t)) cos(th4(t)) sin(th3(t)), -cos(th1(t)) sin(th4(t)),  cos(th1(t)) cos(th3(t))
|
| cos(th1(t)) cos(th3(t)) - cos(th4(t)) sin(th1(t)) sin(th3(t)),  sin(th1(t)) sin(th4(t)), - cos(th1(t)) sin(th3(t)
```

```
|
|                    -sin(th3(t)) sin(th4(t)),                          cos(th4(t)),                           -cos(
|
\                               0,                                           0,

where

   #1 == l2 cos(th2(t)) + l3 cos(th3(t))
Transformación local A5
/  cos(th5(t)),  0, sin(th5(t)),  l5 cos(th5(t)) \
|                                                |
| -sin(th5(t)),  0, cos(th5(t)), -l5 sin(th5(t)) |
|                                                |
|       0,      -1,      0,             0        |
|                                                |
\       0,       0,      0,             1        /
Transformación global T5
/       cos(th5(t)) #2 + cos(th1(t)) sin(th4(t)) sin(th5(t)),      sin(th1(t)) sin(th3(t)) - cos(th1(t)) cos(th3(t)
|
|       cos(th5(t)) #1 - sin(th1(t)) sin(th4(t)) sin(th5(t)),      cos(th1(t)) sin(th3(t)) + cos(th3(t)) cos(th4(t)
|
| - cos(th4(t)) sin(th5(t)) - cos(th5(t)) sin(th3(t)) sin(th4(t)),                   cos(th3(t)) sin(th4(t)),
|
\                                0,                                                               0,

where

   #1 == cos(th1(t)) cos(th3(t)) - cos(th4(t)) sin(th1(t)) sin(th3(t))

   #2 == cos(th3(t)) sin(th1(t)) + cos(th1(t)) cos(th4(t)) sin(th3(t))

   #3 == l2 cos(th2(t)) + l3 cos(th3(t))
```

```
%% Jacobiano diferencial

disp('Jacobiano lineal por derivación directa');
```

```
Jacobiano lineal por derivación directa
```

```
Jv_diff = simplify(jacobian(PO(:,:,GDL),Q));
pretty(Jv_diff)
```

```
/  cos(th1(t)) #4 - l4 sin(th1(t)) sin(th4(t)) + l5 cos(th5(t)) #2 - l5 sin(th1(t)) sin(th4(t)) sin(th5(t)),  -l2 s
|
| - sin(th1(t)) #4 - l4 cos(th1(t)) sin(th4(t)) - l5 cos(th5(t)) #3 - l5 cos(th1(t)) sin(th4(t)) sin(th5(t)), -l2 c
|
\                                                    0,

where

   #1 == l4 cos(th4(t)) + l5 cos(th4(t)) sin(th5(t)) - l5 cos(th5(t)) sin(th3(t)) sin(th4(t))

   #2 == cos(th1(t)) cos(th3(t)) - cos(th4(t)) sin(th1(t)) sin(th3(t))

   #3 == cos(th3(t)) sin(th1(t)) + cos(th1(t)) cos(th4(t)) sin(th3(t))

   #4 == #5 + l3 cos(th3(t))

   #5 == l2 cos(th2(t))
```

```
%% Jacobiano analítico

Jv_a = sym(zeros(3,GDL));
Jw_a = sym(zeros(3,GDL));

for k = 1:GDL
    if RP(k) == 0   % Articulación rotacional
        if k == 1
            z = [0;0;1];
            p = [0;0;0];
        else
            z = RO(:,3,k-1);
            p = PO(:,:,k-1);
        end
        Jv_a(:,k) = cross(z, PO(:,:,GDL) - p);
        Jw_a(:,k) = z;
    else            % Articulación prismática
        if k == 1
            Jv_a(:,k) = [0;0;1];
        else
            Jv_a(:,k) = RO(:,3,k-1);
        end
        Jw_a(:,k) = [0;0;0];
    end
end

Jv_a = simplify(Jv_a);
Jw_a = simplify(Jw_a);

disp('Jacobiano lineal analítico');
```

```
Jacobiano lineal analítico
```

```
pretty(Jv_a)
```

```
/ l4 sin(th1(t)) sin(th4(t)) - cos(th1(t)) (#3 + #2) - l5 cos(th5(t)) (cos(th1(t)) cos(th3(t)) - cos(th4(t)) sin(th
|
| sin(th1(t)) (#3 + #2) + l4 cos(th1(t)) sin(th4(t)) + l5 cos(th5(t)) (cos(th3(t)) sin(th1(t)) + cos(th1(t)) cos(th
|
\                                                                                          0,

where

                     2
   #1 == cos(th4(t))

   #2 == l3 cos(th3(t))

   #3 == l2 cos(th2(t))

   #4 == #8 + l2 sin(th2(t)) + #7 + #6 + #5

   #5 == l5 cos(th5(t)) sin(th3(t)) sin(th4(t))

   #6 == l5 cos(th4(t)) sin(th5(t))
```

```
   #7 == l3 sin(th3(t))

   #8 == l4 cos(th4(t))
```

```
disp('Jacobiano angular analítico');
```

```
Jacobiano angular analítico
```

```
pretty(Jw_a)
```

```
/ 0,      #1,          #1,     -cos(th1(t)) sin(th4(t)),  cos(th1(t)) cos(th3(t)) cos(th4(t)) - sin(th1(t)) sin(th3
|
| 0, sin(th1(t)), sin(th1(t)),  sin(th1(t)) sin(th4(t)), - cos(th1(t)) sin(th3(t)) - cos(th3(t)) cos(th4(t)) sin(th1
|
\ 1,      0,           0,             cos(th4(t)),                           -cos(th3(t)) sin(th4(t))

where

   #1 == -cos(th1(t))
```

```
%% Velocidades del efector final

disp('Velocidad lineal del efector final');
```

```
Velocidad lineal del efector final
```

```
V = simplify(Jv_a*Qp');
pretty(V)
```

```
/ #2 (l5 cos(th1(t)) cos(th5(t)) sin(th4(t)) - l5 cos(th3(t)) sin(th1(t)) sin(th5(t)) + l5 cos(th1(t)) cos(th4(t)) s
|
| #8 (sin(th1(t)) (#7 + #6) + l4 cos(th1(t)) sin(th4(t)) + l5 cos(th5(t)) (cos(th3(t)) sin(th1(t)) + cos(th1(t)) cos
|
\

where

                      2
   #1 == cos(th4(t))

         _________
          d
   #2 == -- th5(t)
         dt

         _________
          d
   #3 == -- th4(t)
         dt

         _________
          d
   #4 == -- th3(t)
         dt

         _________
          d
   #5 == -- th2(t)
```

```
        dt

   #6 == l3 cos(th3(t))

   #7 == l2 cos(th2(t))

         _________
          d
   #8 == -- th1(t)
         dt

   #9 == #13 + l2 sin(th2(t)) + #12 + #11 + #10

   #10 == l5 cos(th5(t)) sin(th3(t)) sin(th4(t))

   #11 == l5 cos(th4(t)) sin(th5(t))

   #12 == l3 sin(th3(t))

   #13 == l4 cos(th4(t))
```

```
disp('Velocidad angular del efector final');
```

```
Velocidad angular del efector final
```

```
W = simplify(Jw_a*Qp');
pretty(W)
```

```
/ - #1 (sin(th1(t)) sin(th3(t)) - cos(th1(t)) cos(th3(t)) cos(th4(t))) - #4 cos(th1(t)) - #3 cos(th1(t)) - #2 cos(th
|
|  #4 sin(th1(t)) - #1 (cos(th1(t)) sin(th3(t)) + cos(th3(t)) cos(th4(t)) sin(th1(t))) + #3 sin(th1(t)) + #2 sin(th
|
|
|                                         _________
|                                          d
|                                         -- th1(t) + #2 cos(th4(t)) - #1 cos(th3(t)) sin(th4(t))
\                                         dt

where

         _________
          d
   #1 == -- th5(t)
         dt

         _________
          d
   #2 == -- th4(t)
         dt

         _________
          d
   #3 == -- th3(t)
         dt

         _________
          d
   #4 == -- th2(t)
         dt
```

```
disp('-----------------------------------------------------------');
```

```
disp('Presiona ENTER para continuar con el robot cartesiano');
```

```
Presiona ENTER para continuar con el robot cartesiano
```

```
pause;

% ------------------------------------------
% ANALISIS CINEMATICO - ROBOT CARTESIANO

clearvars
clc

syms th1(t) th2(t) th3(t) t
syms l1 l2 l3

RP = [0 0 0];
Q  = [th1 th2 th3];
Qp = diff(Q,t);
GDL = length(RP);

disp('Variables articulares del robot cartesiano');
```

```
Variables articulares del robot cartesiano
```

```
pretty(Q)
```

```
(th1(t), th2(t), th3(t))
```

```

%% Definición geométrica

P(:,:,1) = [0; 0; l1];
R(:,:,1) = [0, -sin(th1), -cos(th1);
            0,  cos(th1), -sin(th1);
            1,         0,         0];

P(:,:,2) = [0; 0; l2];
R(:,:,2) = [cos(th2),  0,  sin(th2);
            sin(th2),  0, -cos(th2);
                   0,  1,        0];

P(:,:,3) = [0; 0; l3];
R(:,:,3) = [cos(th3), -sin(th3),   0;
            sin(th3),  cos(th3),   0;
                   0,         0,   1];

%% Transformaciones homogéneas

Vector_Zeros = zeros(1,3);
A = sym(zeros(4,4,GDL));
```

```
T = sym(zeros(4,4,GDL));
PO = sym(zeros(3,1,GDL));
RO = sym(zeros(3,3,GDL));

for i = 1:GDL
    A(:,:,i) = simplify([R(:,:,i) P(:,:,i); Vector_Zeros 1]);

    if i == 1
        T(:,:,i) = A(:,:,i);
    else
        T(:,:,i) = T(:,:,i-1)*A(:,:,i);
    end

    T(:,:,i) = simplify(T(:,:,i));
    RO(:,:,i) = T(1:3,1:3,i);
    PO(:,:,i) = T(1:3,4,i);
end

disp('Jacobiano lineal diferencial');
```

```
Jacobiano lineal diferencial
```

```
pretty(simplify(jacobian(PO(:,:,GDL),Q)))
```

```
/ l2 sin(th1(t)) + l3 cos(th1(t)) cos(th2(t)), -l3 sin(th1(t)) sin(th2(t)), 0 \
|                                                                             |
| l3 cos(th2(t)) sin(th1(t)) - l2 cos(th1(t)),  l3 cos(th1(t)) sin(th2(t)), 0 |
|                                                                             |
\                      0,                             l3 cos(th2(t)),      0 /
```